GENERATIVE AI & AGENTIC AI · LESSON 1 OF 12

How Large Language Models Work: Tokens, Embeddings, Transformers and Training

Open the black box. Tokens and tokenisation (built from scratch), next-token prediction and sampling with temperature, embeddings, attention and the transformer, pre-training, instruction tuning and RLHF, context windows, reasoning models, why LLMs hallucinate, and how to choose among Claude, GPT, Gemini, Llama, Mistral, DeepSeek and Qwen.

DVA + AI class 1 (deep dive) · ✦ 2026 model landscape · Beginner · ⏱ 100 min

By the end of this lesson you can

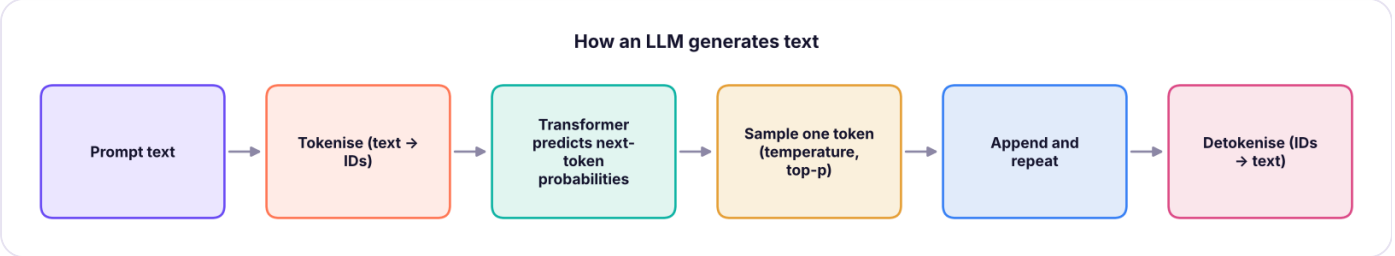
- Explain tokens, vocabularies and why token counts drive cost and limits
- Describe next-token prediction and the effect of temperature and top-p sampling
- Explain embeddings, self-attention and the transformer at an intuitive level
- Describe the training stages (pre-training, instruction tuning, preference optimisation) and reasoning models
- Explain hallucinations and choose a model by capability, cost, latency, privacy and context length

Datasets used

`customer_reviews.csv`

An LLM in one sentence

A large language model is a neural network trained to **predict the next token** given the previous ones. Doing that very well on trillions of words forces it to learn grammar, facts, reasoning patterns and styles, which instruction tuning then turns into a helpful assistant.



Text generation is a loop: tokenise, predict a probability for every possible next token, sample one, append it, repeat.

Tokens

Models do not read characters or whole words but **tokens**: frequent words, word pieces and punctuation. English averages roughly 4 characters (¾ of a word) per token; Indian languages and code usually need more tokens per word.

Python

```python
import re, collections, pandas as pd, numpy as np
corpus = pd.read_csv("customer_reviews.csv").review_text.str.lower().tolist()
def bpe_train(texts, merges=60):
    words = collections.Counter(w for t in texts for w in re.findall(r"[a-z']+", t))
    vocab = {tuple(w) + ("</w>",): c for w, c in words.items()}
    rules = []
    for _ in range(merges):
        pairs = collections.Counter()
        for sym, c in vocab.items():
            for a, b in zip(sym, sym[1:]): pairs[a, b] += c
        best = max(pairs, key=pairs.get); rules.append(best)
        new = {}
        for sym, c in vocab.items():
            out, i = [], 0
            while i < len(sym):
                if i < len(sym) - 1 and (sym[i], sym[i + 1]) == best: out.append(sym[i] +
sym[i + 1]); i += 2
                else: out.append(sym[i]); i += 1
            new[tuple(out)] = c
        vocab = new
    return rules
def bpe_encode(word, rules):
    sym = list(word) + ["</w>"]
    for a, b in rules:
        i = 0
        while i < len(sym) - 1:
            if sym[i] == a and sym[i + 1] == b: sym[i:i + 2] = [a + b]
            else: i += 1
    return sym
rules = bpe_train(corpus, merges=300)
print("first merges learned:", ["".join(r) for r in rules[:10]])
for w in ["battery", "charging", "superb", "unboxing", "skillnavora"]: print(f"{w:12s} →",
bpe_encode(w, rules))
```

OUTPUT

```
first merges learned: ['s</w>', 'e</w>', 'y</w>', 't</w>', 'd</w>', 'er', 'th', 're',
'ar', 'in']
battery      → ['battery</w>']
charging     → ['charg', 'ing</w>']
superb       → ['superb</w>']
unboxing     → ['u', 'n', 'b', 'o', 'x', 'ing</w>']
skillnavora  → ['s', 'k', 'il', 'l', 'n', 'a', 'v', 'or', 'a</w>']
```

This is **byte-pair encoding (BPE)**, the algorithm (in byte-level form) behind GPT, Llama and Claude tokenisers: start from characters and repeatedly merge the most frequent adjacent pair. Common words become one token; rare words split into pieces.

Why tokens matter	Example
Cost	APIs charge per million input and output tokens
Limits	The context window (e.g. 200K–1M tokens) caps prompt + conversation
Speed	Output tokens are generated one at a time
Quirks	Counting letters or reversing words is hard when words are chunks

Count tokens with the provider's tools (Anthropic `messages.count_tokens`, OpenAI `tiktoken`) instead of guessing.

Next-token prediction and sampling

The model outputs a **probability for every token in its vocabulary**. A tiny bigram model trained on our reviews shows the idea:

Python

```
tokens = [w for t in corpus for w in re.findall(r"[a-z']+|[.,!]", t)]
bigram = collections.defaultdict(collections.Counter)
for a, b in zip(tokens, tokens[1:]): bigram[a][b] += 1
def next_probs(word, k=5):
    c = bigram[word]; total = sum(c.values()); return [(w, round(n / total, 3)) for w, n in
c.most_common(k)]
for w in ["battery", "delivery", "not", "worth"]: print(f"after '{w}':", next_probs(w))
```

OUTPUT

```
after 'battery': [('backup', 0.228), ('lasts', 0.174), ('drains', 0.144), ('barely',
0.125), ('swelled', 0.125)]
after 'delivery': [('was', 1.0)]
after 'not': [('worth', 0.283), ('bad', 0.214), ('disappointed', 0.169), ('as',
0.128), ('happy', 0.106)]
after 'worth': [('every', 0.495), ('the', 0.256), ('it', 0.199), ('price', 0.05)]
```

Python

```python
rng = np.random.default_rng(3)
def sample(word, temperature=1.0, top_p=1.0):
    c = bigram[word]; words = list(c); logits = np.log(np.array([c[w] for w in words],
float))
    p = np.exp(logits / temperature); p /= p.sum()
    order = np.argsort(-p); keep = order[: max(1, int(np.searchsorted(np.cumsum(p[order]),
top_p) + 1))]   # nucleus (top-p)
    q = np.zeros_like(p); q[keep] = p[keep]; q /= q.sum()
    return words[rng.choice(len(words), p=q)]
def generate(start, n=10, **kw):
    out = [start]
    for _ in range(n):
        if not bigram[out[-1]]: break
        out.append(sample(out[-1], **kw))
    return " ".join(out)
for t in (0.2, 1.0, 2.0): print(f"temperature {t}:", generate("battery", temperature=t))
```

OUTPUT

```
temperature 0.2: battery backup is crisp even outdoors , not worth every rupee
temperature 1.0: battery easily lasts two days great happy with the hinge broke
temperature 2.0: battery backup is smooth colours on calls ! delivery was totally
```

Setting	Effect	Use
Low temperature (0–0.3)	Picks the most likely tokens: focused, repeatable	Extraction, classification, SQL
Higher temperature (0.7–1.0)	More varied wording	Brainstorming, marketing copy
top-p (nucleus)	Sample only from the smallest set covering p of probability	Cuts off unlikely nonsense

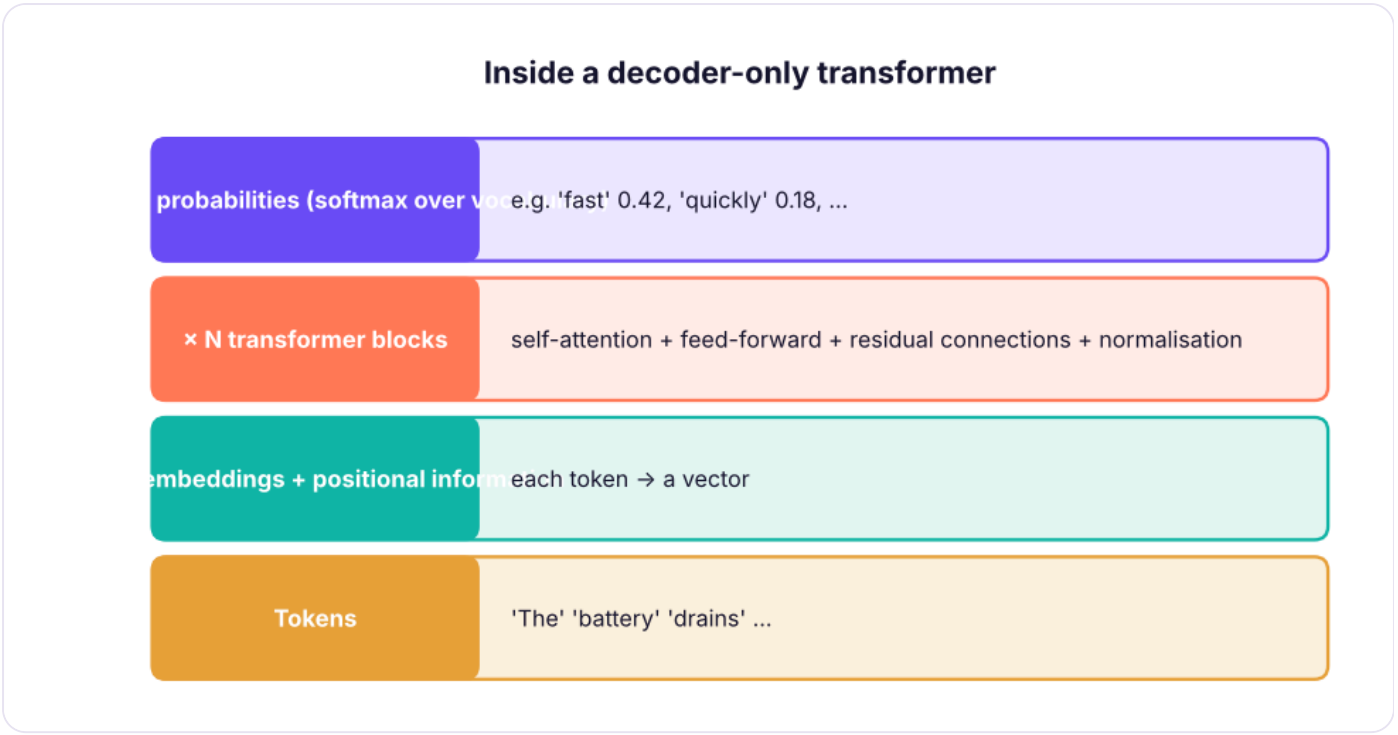
A real LLM does the same thing with a transformer instead of a bigram table, conditioning on the **whole** context, not one word. (Newer reasoning models such as Claude Opus 5.5 manage sampling internally and control depth with an "effort" setting instead.)

Embeddings and attention

Every token ID is mapped to an **embedding**: a vector of hundreds or thousands of numbers. Training places tokens with similar meaning near each other.

Self-attention lets each token look at every other token and decide what matters: in "The battery died because **it** was old", attention links "it" to "battery". A **transformer** stacks dozens of attention +

feed-forward layers; the final layer produces next-token probabilities.



A transformer block repeated N times: attention mixes information across tokens; the feed-forward network transforms each token's representation.

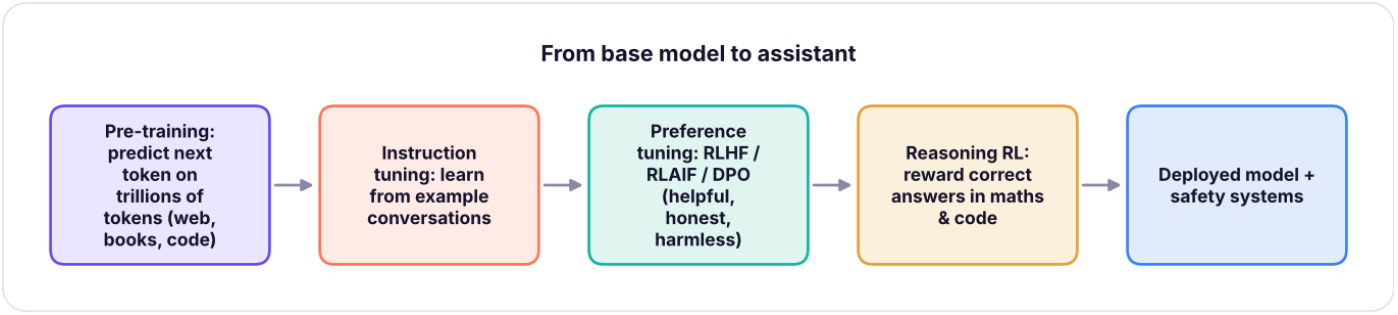
Python

```
def softmax(x): e = np.exp(x - x.max(-1, keepdims=True)); return e / e.sum(-1,
keepdims=True)
words = ["the", "battery", "died", "because", "it", "was", "old"]
rng2 = np.random.default_rng(0); E = rng2.normal(0, 1, (7, 8)); E[4] = E[1] +
rng2.normal(0, .3, 8)      # make "it" resemble "battery"
Q, K = E @ rng2.normal(0, .5, (8, 8)) + E, E
# toy query/key projections
A = softmax(Q @ K.T / np.sqrt(8))
print("attention from 'it' to each word:", dict(zip(words, A[4].round(2).tolist())))
```

OUTPUT

```
attention from 'it' to each word: {'the': 0.01, 'battery': 0.43, 'died': 0.03,
'because': 0.01, 'it': 0.51, 'was': 0.02, 'old': 0.0}
```

How models are trained



The training pipeline of a modern assistant model.

- **Pre-training** produces a *base model* that continues text but does not follow instructions well.
- **Instruction tuning (SFT)** teaches the conversation format.
- **Preference optimisation** (RLHF, Constitutional AI, DPO) aligns answers with human preferences and safety principles.
- **Reasoning models** are further trained with reinforcement learning to "think" before answering (extended or adaptive thinking), which improves maths, coding and planning at the cost of more tokens.

Context windows and memory

An LLM has no memory between API calls: everything it should know must be in the **context window** (system prompt + conversation + documents + tool results). Long contexts (hundreds of thousands to a million tokens) hold whole codebases or reports, but cost more and can dilute attention; retrieval (RAG, lesson 5) puts only relevant passages in context.

Hallucinations

Because the model generates *plausible* continuations, it can state false facts fluently, especially about rare facts, recent events, numbers and citations.

Reduce hallucinations by...	How
Grounding	Provide source documents (RAG) and ask for answers only from them, with citations
Tools	Let the model call search, SQL or a calculator instead of recalling
Prompting	Allow "I don't know"; ask it to quote sources first, then answer
Verification	Check outputs automatically (schemas, SQL execution, tests) and with humans for high stakes
Choice of model	Stronger models hallucinate less; reasoning helps on multi-step problems

The 2026 model landscape

Family	Provider	Notes
Claude (Fable, Opus, Sonnet, Haiku)	Anthropic	Strong reasoning, coding and agentic tool use; 1M-token context on current models
GPT and o-series	OpenAI	Broad ecosystem, multimodal
Gemini	Google	Long context, multimodal, Google Workspace integration
Llama	Meta	Open-weight, self-hostable
Mistral, DeepSeek, Qwen, Gemma	Various	Open-weight options, some very cost-efficient
Sarvam, Krutrim and others	Indian providers	Indic-language focus

Choosing a model: quality on *your* task (test it), cost per task, latency, context length, tool use, data privacy (cloud region, zero retention, self-hosting), and licence. A common pattern: a strong model for hard steps and a small, fast model (e.g. Claude Haiku) for high-volume simple steps.

Summary

KEY TAKEAWAYS

- LLMs predict the next token; generation is a loop of predict, sample, append.
- Tokens (BPE pieces) drive cost, speed and context limits; count them with provider tools.
- Temperature and top-p trade focus for variety; use low randomness for extraction and analytics.
- Embeddings encode meaning; self-attention lets tokens use context; transformers stack attention layers.
- Pre-training → instruction tuning → preference tuning (→ reasoning RL) creates assistants.
- LLMs have no memory beyond the context window and can hallucinate; ground them with documents and tools and verify outputs.

Quick quiz

6 questions · tap an answer to check it

1. What does an LLM fundamentally predict?

A The next token given previous tokens

B The meaning of the whole document at once

C A database query

D Images only

Answer: A. Everything else emerges from that objective.

2. Why do API costs depend on tokens?

- **A** Providers charge per input and output token processed
- **B** Tokens are sold separately
- **C** Only words count
- **D** Costs are fixed per call

Answer: A. Long prompts and long answers cost more.

3. Which temperature suits extracting invoice fields?

- **A** Low (0–0.3)
- **B** High (1.5)
- **C** Temperature does not matter
- **D** Random

Answer: A. Focused, repeatable output.

4. What does self-attention do?

- **A** Lets each token weigh and use information from other tokens in the context
- **B** Removes stop words
- **C** Counts tokens
- **D** Stores long-term memory

Answer: A. e.g. linking "it" to "battery".

5. Why do LLMs hallucinate?

- **A** They generate plausible text, not verified facts, especially for rare or recent information
- **B** They are trained on no data
- **C** Temperature is always 0
- **D** Tokens are too small

Answer: A. Ground and verify.

6. What is RLHF used for?

- **A** Aligning model behaviour with human preferences after pre-training
- **B** Tokenising text
- **C** Increasing the context window
- **D** Compressing images

Answer: A. Reinforcement learning from human feedback.

EXERCISE **Practice**

(1) Train the BPE tokeniser with 50 and 1,000 merges: how many tokens does a 20-word review need in each case? (2) Use the bigram model to generate 5 reviews at temperature 0.7. (3) With the Anthropic token-counting endpoint (GenAI lesson 3), compare token counts of the same sentence in English and Hindi.

Show solution (worked answer)

Python

```
for m in (50, 1000):
    r = bpe_train(corpus, merges=m); review = corpus[5]
    print(m, sum(len(bpe_encode(w, r)) for w in re.findall(r"[a-z']+", review)),
"tokens for", len(review.split()), "words")
for _ in range(5): print(generate("the", n=12, temperature=0.7))
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain how ChatGPT or Claude works to a business audience. It is a model trained on vast text to predict the next word-piece; that teaches it language and knowledge. Further training with example conversations and human feedback makes it follow instructions helpfully and safely. It does not look things up unless connected to tools or documents.

Q: What is a context window? The maximum number of tokens (prompt, history, documents, output) the model can consider in one request.

Q: How do you reduce hallucinations in a business application? Retrieval-augmented generation with citations, tool use for facts and numbers, instructions to say "I don't know", structured outputs with validation, evaluation sets and human review for high-stakes outputs.

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